CNC Procedure 2016_028

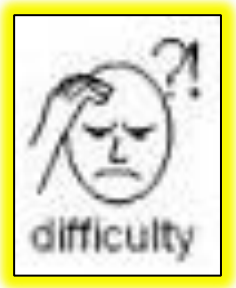
Aug 3rd

2016

Procedure for Setting up hardware for online monitor Servo/Spindle Signals through oscilloscope

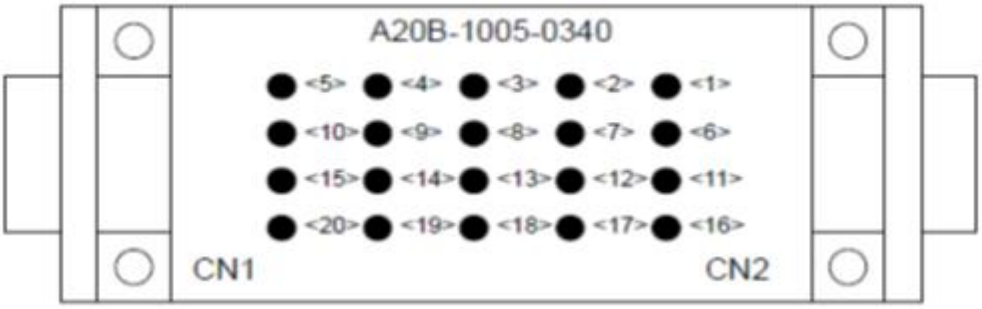
[Applicable for all Alpha/Beta-iA/B Series Amplifiers]

How to check Servo/Spindle Signal Online!!



What are the instruments used for checking?

1. **Oscilloscope → >2 Channel, >100MHz**
2. **Cable for servo check pin board → A660-2042-T031#L200R0**
3. **Servo check pin board BOA → A20B-1005-0340, Pin layout as below:**



4. **Snapshot of instruments used:**

Oscilloscope	(A06B-6071-K290) Connection Cable & PCB

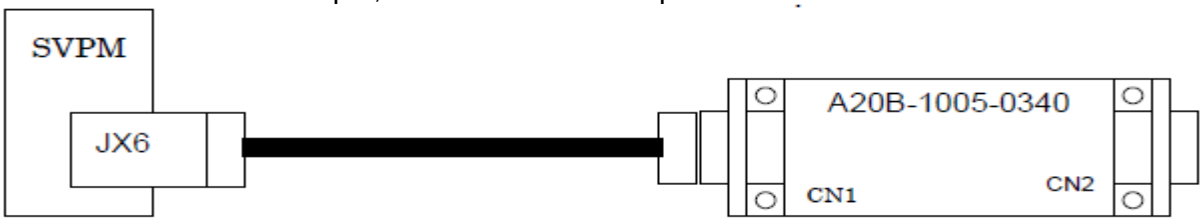
What can be checked online?

1. **Servo/Spindle feedback signals PA, PB, Z, etc.,**
2. **Inter Amplifier communication status like Control Voltage, VRDY, MRDY signal**
3. **Input current (IR/IS) of βi or αi series amplifiers**
4. **Differential amplified signal ADIF, BDIF, PZ & T0 signal levels of external feedback devices like Rotary Encoder/Linear scale**

How to check input current in αi/βi Amplifiers?

βi-SVSP

- Connect cable to connector JX6
- To observe the output, use the servo check pin board A06B-6071-K290 as shown below:



Check pin Description Location of observation Remark

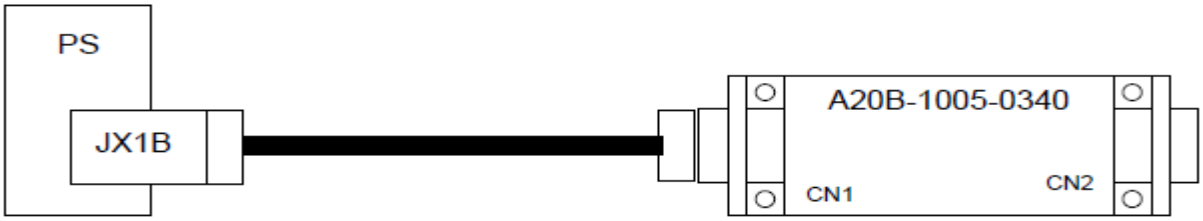
Check	Description	Check Pin	Remark
IR	L1ϕ (Rϕ) Current	JX6-Pin1	The "+" sign with respect to the input of the amplifier. If the L1 or L2 phase current exceeds the overcurrent alarm level, an alarm condition occurs in the βiSVSP
IS	L2ϕ (Sϕ) Current	JX6-Pin2	
0V	Reference point	JX6 – Pin 12/14/16	

IR and IS current conversion value

βiSVSP*-5.5	133A/1V (2.5 V at the center)
βiSVSP*-7.5	133A/1V (2.5 V at the center)
βiSVSP*-11	133A/1V (2.5 V at the center)
βiSVSP*-15	200A/1V (2.5 V at the center)
βiSVSP*-18	200A/1V (2.5 V at the center)

αi-PS

- To the Power Supply connector JX1B, the input current check signal is output.
- To observe the output, use the servo check pin board A06B-6071-K290 as shown below:



Check pin Description Location of observation Remark

Check	Description	Check Pin	Remark
IR	L1ϕ (Rϕ) Current	JX1B-Pin1	The "+" sign with respect to the input of the amplifier. If the L1 or L2 phase current exceeds the overcurrent alarm level, an alarm condition occurs in the βiSVSP
IS	L2ϕ (Sϕ) Current	JX1B-Pin2	
0V	Reference point	JX1B – Pin 12/14/16	

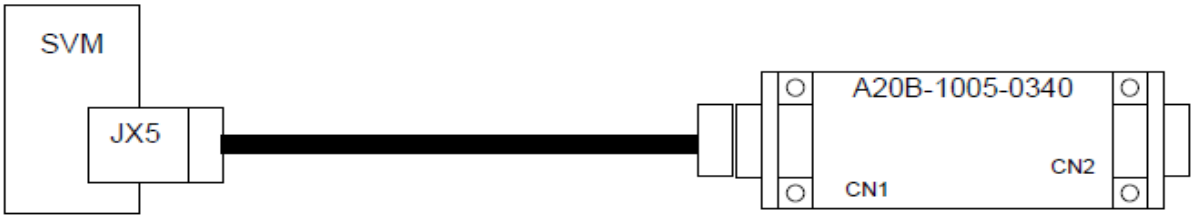
IR and IS current conversion value

Model	Current Value
αiPS-5.5	133A/1V (2.5 V at the center)
αiPS-11	133A/1V (2.5 V at the center)
αiPS 15	200A/1V (2.5 V at the center)
αiPS 26	266A/1V (2.5 V at the center)
αiPS 30	333A/1V (2.5 V at the center)
αiPS 37	400A/1V(2.5 V at the center)
αiPS 55	666A/1V(2.5 V at the center)

Model	Current Value
αiPS-11HV	100A/1V (2.5 V at the center)
αiPS-18HV	133A/1V (2.5 V at the center)
αiPS 30HV	200A/1V (2.5 V at the center)
αiPS 45HV	266A/1V (2.5 V at the center)
αiPS 60HV	333A/1V (2.5 V at the center)
αiPS 75HV	400A/1V(2.5 V at the center)
αiPS 100HV	666A/1V(2.5 V at the center)

How to check Spindle Feedback Sensor Signal Waveform in αi/βi Amplifiers?

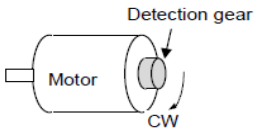
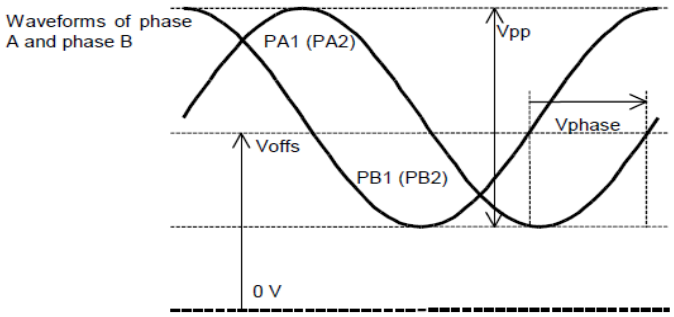
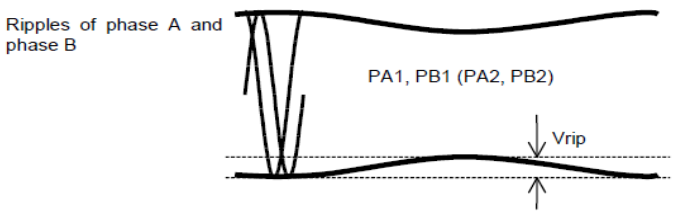
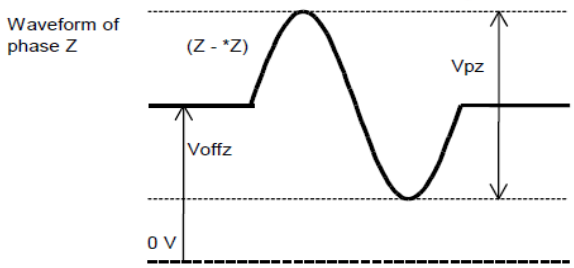
- The feedback wave form measurement connector/pins may vary depending on the configuration of the detector



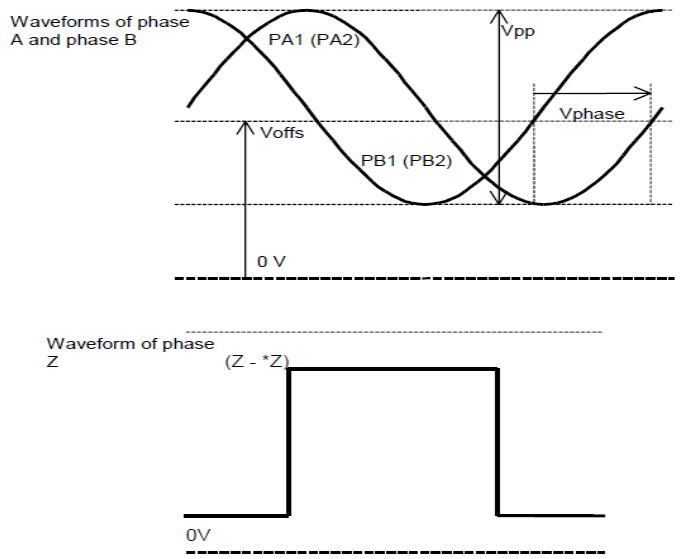
- To check the waveform with below table reference by using the check terminals pins on the check board.

Check Signal Name	Connector Name Pin Number	Sensor info
PA1 **PB1**	JYA2 (Pin5,6) JYA2 (Pin7,8)	αi M, αi MZ, and αi BZ sensors Analog αi CZ sensor
PA2 **PB2**	JYA4 (Pin5,6) JYA4 (Pin7,8)	αi BZ sensor Analog αi CZ sensor α position coder S(1024λ)
PS1	JYA2 (Pin1,2)	αi MZ and αi BZ sensors (one-rotation signal) Analog αi CZ sensor (one-rotation signal)
PS2	JYA4 (Pin1,2)	αi BZ sensor (one-rotation signal) Analog αi CZ sensor (one-rotation signal)
EXTSC1	JYA3 (Pin15)	Proximity switch (external one-rotation signal)

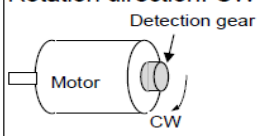
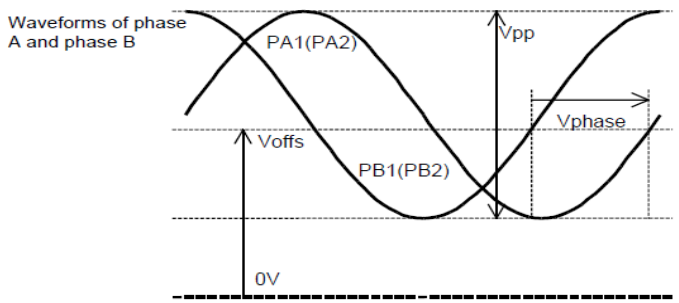
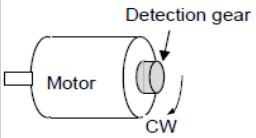
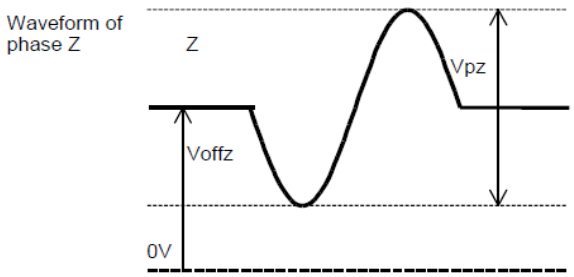
Feedback signal levels αi M sensor, αi MZ sensor, and αi BZ sensor

Measurement location	Measurement condition	Sample waveform
PA1, PB1 Separate sensors PA2, PB2	The speed must be 1500 min^{-1} or less. Rotation direction: CW 	
		
PS1 Separate sensor PS2		For αi MZ and αi BZ sensors only 

Feedback signal levels α - Position Coder

Measurement location	Measurement condition	Sample waveform
PA2, PB2	CW rotation direction as viewed from the flange	

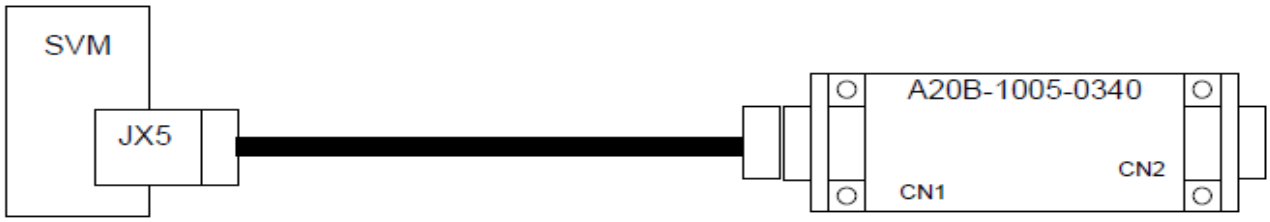
Feedback signal levels α i-CZ Sensor

Measurement location	Measurement condition	Sample waveform
PA1, PB1 Separate sensors PA2, PB2	Speed: 500min^{-1} Rotation direction: CW 	
PS1 Separate sensor PS2	Speed: 500min^{-1} Rotation direction: CW 	

Feedback signal levels for above sample waveforms

Measurement item	Specification	Measurement method	Adjustment method
Vpp (for αìM/MZ/BZ)	0.5~1.2 Vp-p	Oscilloscope in DC mode	Normally, only the α*i* M and α*i* MZ sensors need not be adjusted. For Voffs and Voffz, only level check is possible, but adjustment is not possible.
Vpp (for αìCZ)	0.9~1.2 Vp-p		
Vpp (for αPositoin coder)	0.8~1.2 Vp-p		
Voffs, Voffsz	2.5V±100mV		
Vphase	90±3^{0}	Oscilloscope in duty mode	
Vrip	<70mV		
Vpz (for αìM/MZ/BZ)	>0.5V`		
Vpz (for αìCZ)	0.66 to 1.65V p-p		

α-Series – SVM:



Signal Output: SVM designed for operation with the FSSB interface

1. Do not touch the clear signal of the control circuit, otherwise the servo amplifier may malfunction.
2. Part of the signal may not be output depending on the revision of the printed circuit board.
3. Pin, Various Signal with voltage levels are listed as below:

Pin No.	Signal name			Remarks
	SVM1	SVM2	SVM3	
<1>	CLR	CLR	CLR	Clear signal of the control circuit(Note)
<2>	+3.3V	+3.3V	+3.3V	+3.3-V power (Variations in voltage level within ±5% are regarded as normal.)
<3>	IRL	IRL	IRL	Phase R motor current for the L axis
<4>	ISL	ISL	ISL	Phase S motor current for the L axis
<5>		IRM	IRM	Phase R motor current for the M axis
<6>		ISM	ISM	Phase S motor current for the M axis
<7>	0V	0V		Reference voltage
			IRN	Phase R motor current for the N axis
<8>	0V	0V		Reference voltage
			ISN	Phase S motor current for the N axis
<9>	0V	0V	0V	Reference voltage
<10>				
<11>	+24V	+24V	+24V	+24-V power
<12>	+15V	+15V	+15V	+15-V power
<13>	-15V	-15V	-15V	-15-V power
<14>	+5V	+5V	+5V	+5-V power
<15>	ITL	ITL	ITL	Phase T motor current for the L axis
<16>				
<17>		ITM	ITM	Phase T motor current for the M axis
<18>				
<19>	0V	0V		Reference voltage
			ITN	Phase T motor current for the N axis
<20>				

SVM designed for operation with the type A/B interface

Pin No.	Signal name			Remarks
	SVM1	SVM2	SVM3	
<1>	CLR	CLR	CLR	Clear signal of the control circuit(Note 1)
<2>	0V	0V	0V	Reference voltage
<3>	IRL	IRL	IRL	Phase R motor current for the L axis
<4>	ISL	ISL	ISL	Phase S motor current for the L axis
<5>		IRM	IRM	Phase R motor current for the M axis
<6>		ISM	ISM	Phase S motor current for the M axis
<7>	0V	0V		Reference voltage
			IRN	Phase R motor current for the N axis
<8>	0V	0V		Reference voltage
			ISN	Phase S motor current for the N axis
<9>	0V	0V	0V	Reference voltage
<10>	0V	0V	0V	Reference voltage
<11>	+24V	+24V	+24V	+24-V power
<12>	+15V	+15V	+15V	+15-V power
<13>	-15V	-15V	-15V	-15-V power
<14>	+5V	+5V	+5V	+5-V power
<15>	ITL	ITL	ITL	Phase T motor current for the L axis
<16>				
<17>		ITM	ITM	Phase T motor current for the M axis
<18>				
<19>	0V	0V		Reference voltage
			ITN	Phase T motor current for the N axis
<20>				

Now save all the collected data to one folder along with information of Drive & Controller name plate for future troubleshooting

FOR MORE INFORMATION ON THE ABOVE BRIEFED TOPICS PLEASE REFER RESPECTIVE AMPLIFIER MAINTENACE MANUALS